HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and validated **Python scripts** for automated data capture and analysis in a lab environment, supporting hardware measurements for RF product ICs.
- Performed hands-on silicon evaluation and device measurements using **Spectrum Analyzers**, **Phase Noise Analyzers**, and **Vector Network Analyzers**.
- Contributed to the development of technical training materials for field application engineers, focusing on RF product portfolios including **PLLs**, **VCOs**, and **JESD204**.
- Authored technical documentation including **user guides**, **application notes**, and datasheets for new and existing RF products.
- Collaborated with cross-functional teams to define requirements for new RF products, translating customer application needs into technical specifications for the product definition phase.
- Resolved challenging customer design problems through expert analysis and timely response to technical inquiries from customers and field application engineers.
- Containerized data capture and analysis scripts using **Docker** for reproducible execution in development and testing environments.

SKILLS

Core RF ICs – RF Phase Locked Loops (PLL), Voltage Controlled Oscillators (VCO), integrated PLL/VCOs, JESD204, Network Clock ICs, frequency multipliers, Direct Digital Synthesizers (DDS), frequency dividers

Programming & Scripting – Python, C, C++, C#

Test & Measurement – Spectrum Analyzers, Phase Noise Analyzers, Vector Network Analyzers, Silicon Evaluation

Documentation & Training – Datasheets, User Guides, Application Notes, Technical Training Material Development

Development Tools – Docker, Git, Jenkins

Hardware & Systems – Hardware Design, Block Diagram Verification, FPGA, SoC

PROJECTS

RF Product Evaluation and Data Automation (HCL Technologies)

- Developed a **Python framework** to automate data acquisition from RF test equipment, reducing manual measurement time by 40% for characterization reports.
- Implemented **data validation** routines within the Python scripts to ensure accuracy and consistency of captured device parameters, flagging anomalies for review.
- Designed and executed test plans for silicon evaluation of new RF components, documenting performance against specifications.

Customer Design-In Support for RF Signal Chain (HCL Technologies)

- Assisted enterprise customers with integrating **RF components** into their system designs, providing technical guidance and resolving implementation challenges.
- Created detailed **application notes** and example schematics to demonstrate best practices for using RF PLL and VCO ICs in complex signal chain solutions.
- Supported customer design-in activities by analyzing their application requirements and mapping them to appropriate product specifications.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)